The Approach

Requirements

Functional

- users can create & update an account
- users can login
- Users can search product
- USER can add && remove product from the cart
- User can submite the order
- User can track the order
- system receives orders
- system validates the order
- system makes sure the same order is never processed twice
- system checks that every item is available and reserves it.
- system prevent overselling
- system decides how many robots are needed to collect
- system returns a list of robots with the items assigned to each one
- The system sends pick tasks to the assigned robots and follows their progress
- system must handle robot failures
- if order is fully collected he system moves the order to the packing station and then to shipping
- system notifies operators about low stock, a robot failure, a delayed order, or a robot with a low battery

Non-Functional

1. low latency : system must process validation the order with rsponse time of 100ms at least 98% of request
2. strong consistency : strict ACID in complain immediate database replacementare required for validation the order

3. eventaul consistency : asyncrounce synchronization is allowed for decide robot that select within maximum windows of4 secondsc to priotize high avialbiltiy and read speed

API

- POST /auth/ register ⟶ user-Id

{ name, email , password}
- POST/User/login ⟶token{

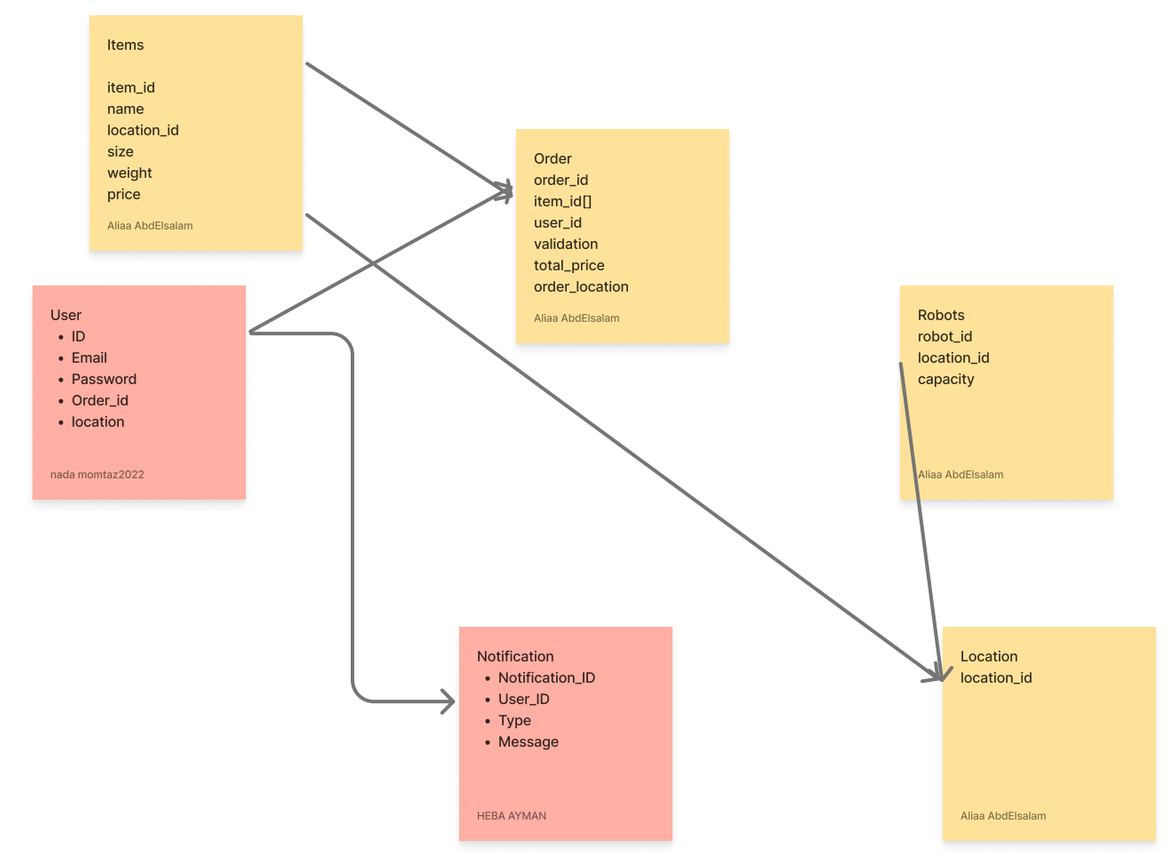
email, password

}
- Get/ Items ⟶ Items[]{id_items , name, price }
- Post/Items/ ⟶ item_id {Name ,price , size , weight }
- PATCH/Cart/ ⟶ 200{}
- POST/Orders ⟶OrderId{
  id_items [] , *location*_user }
- Get / Order/order_id⟶ { order_location}
- Get/order/ {order id}/ ⟶id_items []{}
- Post/order/ {order id}/ user⟶409{order_Id}
- Post /Robots/{robot_id} → 200
  {item_id}

Core Entities

- User
- Order
- Items
- Location
- Notification
- Robots

Data Model


Items
item_id
name
location_id
size
weight
price
Aliaa AbdElsalam
Order
order_id
item_id[]
user_id
validation
total_price
order_location
Aliaa AbdElsalam
User
• ID
• Email
• Password
• Order_id
• location
nada momtaz2022
Robots
robot_id
location_id
capacity
Aliaa AbdElsalam
Notification
• Notification_ID
• User_ID
• Type
• Message
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Location
location_id
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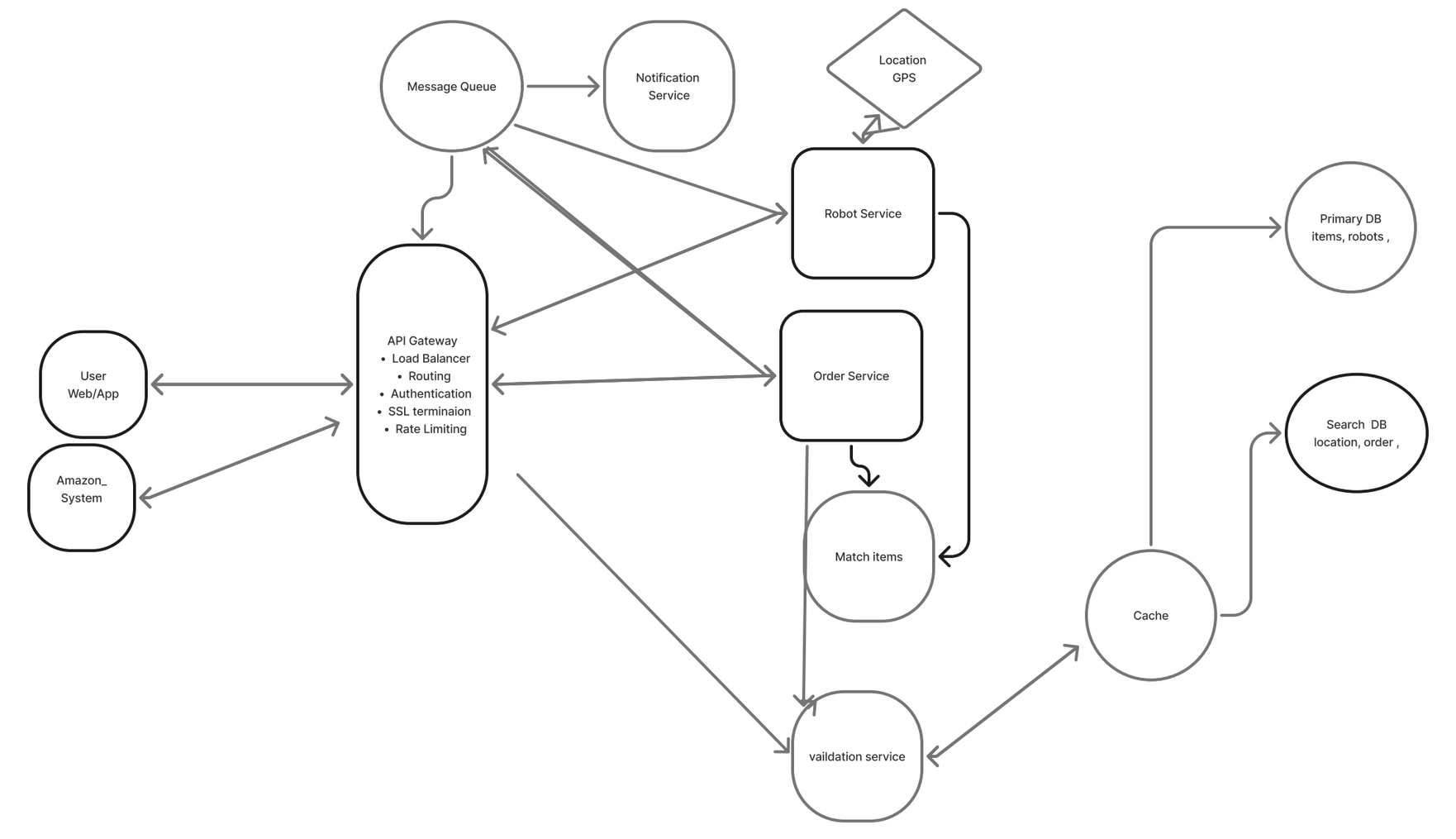
Message Queue
Notification Service
Location GPS
Robot Service
Order Service
Match items
API Gateway
• Load Balancer
• Routing
• Authentication
• SSL terminaion
• Rate Limiting
User Web/App
Amazon_ System
vaildation service
Cache
Primary DB items, robots ,
Search DB location, order ,